NeurIPS 2023 | The First Open Triad for Financial LLMs: Model, Data, and Benchmark

PIXIU introduces FinMA, FIT, and FLARE, showing that domain instruction tuning lets a small open model beat far larger general systems on financial NLP

Financial text is dense with jargon and sensitive to context, and it has long been assumed to need models built for the domain. NLP has moved into finance quickly over the past few years, yet an open financial large language model you could actually build on barely existed. The financial pre-trained models that were available, such as finBERT and FLANG, sit below 1B parameters, which limits how well they generalize. The one large financial model, the 50B BloombergGPT, is fully proprietary: it releases neither its weights nor its training data, and it does not follow instructions.

The deeper problem is that a model alone is not enough. To train and fairly evaluate a large model in finance, you need two more things: a financial instruction-tuning dataset to give the model zero-shot ability, and an evaluation benchmark that covers realistic tasks so different models can be compared head to head. None of this existed in open form, so the research community was effectively locked out of open progress in the field.

PIXIU, from Wuhan University, Sun Yat-Sen University, Southwest Jiaotong University, the University of Florida, and ChanceFocus and published at NeurIPS 2023, fills all three gaps at once: the FIT instruction dataset, the instruction-following financial LLM FinMA, and the FLARE evaluation benchmark. The overview below shows how the three pieces fit together.

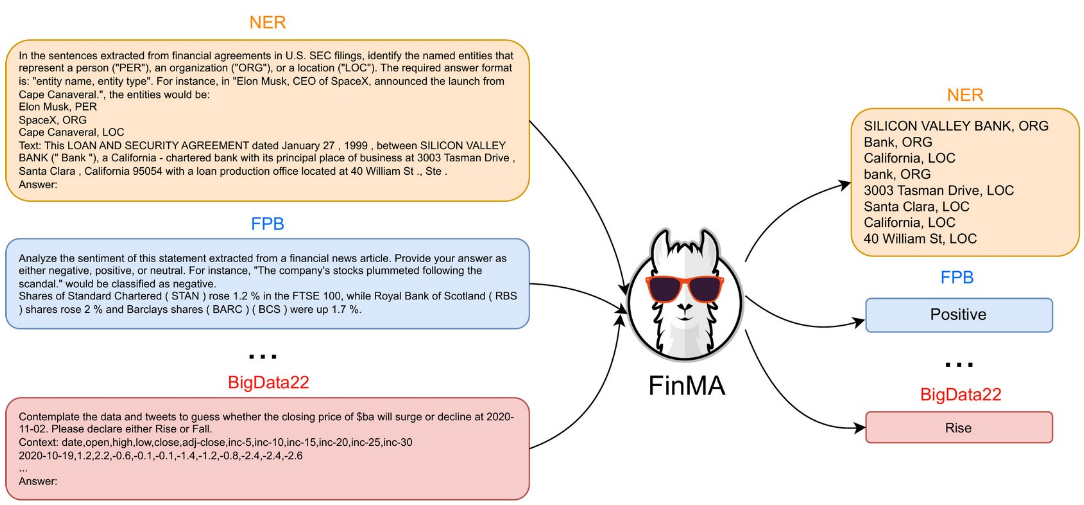

Figure 1. An overview of FinMA's multi-task, multi-modal instruction tuning. On the left are instruction-wrapped inputs for tasks such as named entity recognition (NER), sentiment analysis (FPB), and stock movement prediction (BigData22); on the right is the structured output FinMA produces for each. A single model covers financial tasks that span text, tables, and time series.

Paper: https://arxiv.org/abs/2306.05443 . Code and data: https://github.com/chancefocus/PIXIU . The model, instruction data, benchmark datasets, and implementation are all open-sourced.

The problem: why financial AI lacked an open foundation

Instruction tuning has proven to be a key step for improving a model's zero-shot ability, but with no financial instruction data available, that step was simply out of reach. At the same time, existing financial benchmarks such as FLUE and BBT-CFLEB cover only financial NLP tasks. They leave out stock movement prediction, the kind of task that is closest to a real trading setting and that requires combining text with time-series data.

The community was therefore facing a triple gap: no open model, no open instruction data, and no unified benchmark that included prediction tasks. Any one of these missing pieces makes financial LLM research hard to reproduce and hard to compare. PIXIU sets out to close all three at once, with open resources, multi-task coverage, and multi-modal data.

The method: FIT, FinMA, and FLARE

The first step is the data. PIXIU collects open-released data across five financial tasks: sentiment analysis, news headline classification, named entity recognition, question answering, and stock movement prediction. Domain experts then write diverse task-specific instructions for each task, which are assembled with the samples into the FIT instruction dataset. FIT holds 136,609 instruction samples (about 136K) spanning 5 tasks and 9 datasets, and it is multi-modal, mixing text and tables from financial reports with historical stock prices as time series.

The second step is the model. The authors run multi-task instruction tuning on the open LLaMA checkpoints, producing FinMA at 7B and 30B, plus a FinMA-7B-full variant trained on both NLP and prediction tasks. The third step is evaluation: the FLARE benchmark places four financial NLP tasks (six datasets) alongside one financial prediction task (three datasets), so FinMA and other LLMs can be measured on the same yardstick.

The result: how a small open model beats GPT-4

On FLARE, the instruction-tuned FinMA significantly outperforms other large models on most financial NLP tasks. On the FPB sentiment dataset, for example, FinMA-30B beats GPT-4 by 10% F1 and BloombergGPT by 37% F1; on headline classification and named entity recognition it again leads or matches GPT-4 and ChatGPT. The reading is straightforward: aligning a large model to finance through instruction tuning does more than simply scaling parameters.

Table 1. Representative results on the FLARE benchmark (higher is better; "-" means the original paper did not report a value). FinMA leads on sentiment, headline classification, and entity recognition, but trails on ConvFinQA, which demands numeric reasoning.

Task (metric)	BloombergGPT	GPT-4	FinMA-7B	FinMA-30B
FPB (F1)	0.51	0.78	0.86	0.88
FiQA-SA (F1)	0.75	-	0.84	0.87

Headline (Avg F1)	0.82	0.77	0.98	0.97
NER (Entity F1)	0.61	0.77	0.75	0.62
ConvFinQA (EM Acc)	0.43	0.60	0.25	0.40

There is an honest flip side. On question answering that demands complex numeric reasoning, such as FinQA and ConvFinQA, FinMA falls clearly behind the GPT family. The authors trace this to the backbone: LLaMA saw almost no mathematical data during pre-training, so its numeric reasoning is weak to begin with, and instruction tuning cannot conjure that ability from nothing. Domain alignment fixes language understanding, but it does not substitute for reasoning the base model never had.

Scale is not the answer: what the ablations show

Growing FinMA from 7B to 30B brings no consistent gain on most NLP tasks or on stock prediction. This suggests that, for financial instruction tuning, the quality and diversity of the data matter more than raw parameter count. Scale helps only on strongly numeric question answering like ConvFinQA, tracking LLaMA's improving math ability at larger sizes, though it still trails GPT-4.

A second observation comes from FinMA-7B-full, which is instruction-tuned on both NLP and prediction tasks: it reaches the best score of any model on the ACL18 stock prediction dataset. This hints that task-specific instruction tuning for prediction may be a viable path toward genuinely useful financial forecasting, even though stock movement prediction remains hard for every model in the study.

Takeaway: an open starting point that also marks its limits

PIXIU matters because it opens up the full financial-AI triad for the first time: an instruction-following financial LLM in FinMA, a large multi-task instruction dataset in FIT, and a holistic benchmark spanning language and prediction in FLARE. It shows empirically that careful domain instruction tuning can let a relatively small open model outperform much larger general-purpose systems on financial language tasks.

At the same time, the paper does not paper over the boundary. Numeric-reasoning question answering, and stock movement prediction that stays difficult across every model, are marked clearly as the field's open problems. For anyone looking to study or deploy large models in finance, what PIXIU offers is a reproducible, comparable, and extendable open foundation to build on.